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POPULATION DYNAMICS BASED ON SIZE OR AGE? A REEF-CORAL ANALYSIS

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Biologists have long realized that individuals in a population often have dissimilar expectations of growing, reproducing, or dying and that much of this variation can be attributed to differences in age or size (e.g., Deevey 1947; Connell 1973; Harper 1977; Hughes and Jackson 1985; Caswell 1987). Population studies were first applied to solitary animals that develop determinately through a one-way sequence of life stages, that reach a maximum adult size determined in part by genetics, and that eventually senesce. The dynamics of these organisms may be well predicted by models incorporating age-dependent rates of birth, death, and migration (Nisbet and Gurney 1982).

However, other organisms such as most sessile, clonal animals and many perennial plants possess none of the above traits and, therefore, often have a much more plastic development. In these organisms, variable rates of growth, fission, fusion, and in many cases great longevity combine to produce a wide range of sizes and demographic rates among individuals within a single cohort (Werner 1975; Hughes and Jackson 1980, 1985; Hughes 1984). To cope with such large amounts of variability within an age class, a number of recent population models have been based on size or developmental stage, rather than age (e.g., Lefkovitch 1965; Usher 1972; Sarukhán and Gadgil 1974; Werner and Caswell 1977; Caswell and Werner 1978; Hubbell and Werner 1979; Law 1983; Hughes 1984; Kirkpatrick 1984; Caswell 1985). Thus, population biologists have traditionally viewed age and size as alternative ways of classifying populations into demographically useful categories, the choice of one or the other depending on the organism's mode of development and life history and on the amount of information available. Here we demonstrate, for the first time for an animal population, simultaneous effects of both size and age on the growth and mortality of reef corals.

The mode of construction of clonal organisms such as colonial corals greatly increases the complexity of their life histories and population dynamics, compared with those of more-familiar aclonal (solitary) animals. A coral colony is composed of interconnected building blocks, polyps, that are derived by vegetative (asexual) budding from a common larval ancestor. Colonies grow primarily by

the addition of new building blocks, and not by continually increasing the size of each polyp. Although the size of polyps, like that of individual aclonal organisms, is genetically determined, growth of the colony as a whole is potentially indeterminate (Jackson 1979; Jackson and Hughes 1985). However, because local environments experienced by sessile colonies often differ greatly, the inherent capacity for growth may be extrinsically reduced by amounts varying in space and time (e.g., by local physical conditions or distance to neighbors; Connell 1973; Hughes 1984). Colonial corals are frequently damaged by many agents; but by definition colonies possess replicated organs within each polyp, and they therefore have considerable powers of regeneration and can survive extensive losses of tissue and skeleton (Connell 1973; Hughes and Jackson 1980; Bak 1983). In many species, especially those with a loosely branched morphology, damaged corals may break up to form several daughter colonies (Highsmith 1982). These traits, differential growth rates, shrinkage, and fission of colonies, all tend to decouple the relationship between a colony's age and its size (Hughes 1984).

Many aspects of the dynamics of coral population have recently been related to colony size. For example, relative rates of growth in colony area decrease with increasing colony size (Connell 1973; Hughes and Jackson 1985). Similarly, whole-colony mortality also declines sharply among larger colonies (see, e.g., Connell 1973; Pearson 1974; Glynn 1976; Loya 1976; Highsmith et al. 1980). In contrast, rates of partial mortality (shrinkage) and fission of colonies increase with size (Hughes and Jackson 1980, 1985; Hughes 1984), as does the fecundity of the colonies (Connell 1973; Rinkevich and Loya 1979; Kojis and Quinn 1981, 1985; Van Moorsel 1983; Babcock 1984; Szmant-Froelich 1985). Much less is known concerning the role of colony age (Connell 1973; Hughes 1985) because of the difficulties of accurately estimating it without damaging the colony (Hughes and Jackson 1980).

In this paper, we quantify demographic rates for coral colonies of known age and size that were observed repeatedly over a period of 8 yr. We show that size effects are stronger than age effects but that colony age often has a significant influence on dynamics, even within a single size class.

METHODS

Growth and mortality of 405 coral colonies of known age and size were measured from 1970 to 1971 on the reef crest at Heron Island, Queensland, Australia, 23°S. A series of photographs of 11 permanently marked, 1-m quadrats was taken in July or August of 1963, 1965, 1969, 1970, and 1971 and in February 1967. Using these photographs, we were able to follow the individual colonies and estimate coral age. Taxonomic identifications were made by C. C. Wallace, from either collected specimens or photographs. Color slides from each census were projected onto a screen at 0.5 magnification and traced. Colonies were then numbered and followed through time; the survivors in 1970 ranged in age from less than 1 yr to 6 or more years. Relative growth of colonies, defined as the change in projected colony area divided by the initial area, was measured from 1970 to 1971 by planimetry. Since the corals are covered by only a few centimeters of water at low

TABLE 1
AGE AND SIZE DISTRIBUTION OF CORALS IN 1970

Size	<i>Acropora</i>			<i>Porites</i>			<i>Pocillopora</i>		
	Young	Old	Total	Young	Old	Total	Young	Old	Total
0–10 cm ²	35	34	69	34	23	57	29	9	38
10–50 cm ²	39	40	79	19	11	30	29	10	39
>50 cm ²	13	69	82	0	3	3	4	4	8
Total	87	143	230	53	37	90	62	23	85

NOTE.—Old colonies were first seen in 1963 or 1965, young colonies in 1967, 1969, or 1970.

tide, most of their growth occurs horizontally. Therefore, changes in projected area provide an adequate measure of growth. If corals underwent fission during that year, areas of daughter colonies were combined. Mortality of whole colonies was observed directly by the disappearance of all live tissues. The resolution of the photographs allows detection of colonies of 1 cm² or greater; these are aged approximately 8–12 mo after settlement (Wallace 1983; Babcock 1985). Thus, our data, like those from most studies of marine invertebrates (Connell 1985), do not address the fate of post-settlement corals before they are detected as recruits.

Virtually all of the corals were *Acropora* (11 species), *Porites* (3 species), or *Pocillopora damicornis*. The differences in life histories are far greater between the above groups than within each category (Connell 1973; Connell and Hughes, unpubl. data). We therefore grouped colonies into these three taxa in order to obtain large sample sizes. Apart from the taxonomic differences, the three genera also represent distinct morphological groups: on the reef crest, *Acropora* range from encrusting to branching, species of *Porites* form small flat mounds, and colonies of *Pocillopora damicornis* form tightly branching bushes. Other corals that were less common within the permanent study areas (e.g., species of *Montipora*, *Favia*, *Goniastrea*, and *Stylophora*) were not included in the present analysis.

By 1970, many of the extant colonies had recruited since 1963. Therefore, they could be assigned to age classes, up to 6 or more years old, on the basis of the time since one of the six censuses when they were first observed. In all of our statistical analyses we divided the corals in 1970 into two age categories, older and younger than 4.5 yr, in order to minimize any artifacts of limited photographic resolution and to give equal numbers of old and young colonies (202 young, 203 old).

Because a hurricane caused extensive damage on Heron Island after the 1971 census, coral populations subsequently observed (until the present day) often had only a small number of old colonies. Thus, the choice of 1970–1971 as the period over which to test for size and age effects was based only on constraints of sampling.

RESULTS

The age-class and size-class distributions of the three taxa in 1970 are summarized in table 1. The data show clearly that virtually every size class contains both young and old colonies. This pattern allows us to describe rates of mortality

TABLE 2
SIZE-SPECIFIC AND AGE-SPECIFIC MORTALITY

1970	DIED (%)			ALL SPECIES
	<i>Acropora</i>	<i>Porites</i>	<i>Pocillopora</i>	
A. Size-Specific Mortality				
Initial size				
0–10 cm ²	46	51	39	46
10–50 cm ²	27	10	8	18
>50 cm ²	4	0	0	3
χ ² value	37.5***	16.1***	14.0***	64.7***
B. Age-Specific Mortality				
Initial age				
Young	28	38	18	27
Old	22	32	30	25
χ ² value	1.0	0.3	1.6	0.4

NOTE.—See table 1 for sample sizes.

*** $P < 0.001$.

and net growth from 1970 to 1971 as a function of size regardless of age, as a function of age regardless of size, and as a function of age within a single size class.

Mortality

Overall, 26% of the colonies died between 1970 and 1971. Mortality depended greatly on colony area, decreasing fifteenfold among the three size classes (table 2A). Almost half of all corals smaller than 10 cm² died, compared to a loss of less than 20% for colonies between 10 cm² and 50 cm², and of only 3% for corals larger than 50 cm². The size effect was highly significant for all three taxa. In contrast, we found no significant effect of age on mortality (table 2B). Taken together, the data show that 25% of old colonies (older than 4.5 yr) were killed but that almost all larger than 50 cm² survived. Thus, most of the old colonies that died were also small.

Growth

Relative growth of colonies also correlated strongly with initial size (fig. 1); about half of colonies smaller than 10 cm² in 1970 grew to at least double their initial area (100% growth per year) by 1971, and many increased threefold or more. In contrast, all but three corals larger than 100 cm² grew by less than 100%, and almost half of them suffered a net loss of tissue (fig. 1). The decline in net relative growth with increasing colony size is to be expected, since an expanding colony would have to add on new periphery at an ever-increasing pace to maintain a constant rate of growth. In addition, we show below that the relationship between relative growth and size, caused in part by this geometric constraint, varies further as a function of age.

Relative growth also declined with increasing age, regardless of initial size (fig. 2). We rejected the hypothesis of equal growth rates for old and young colonies

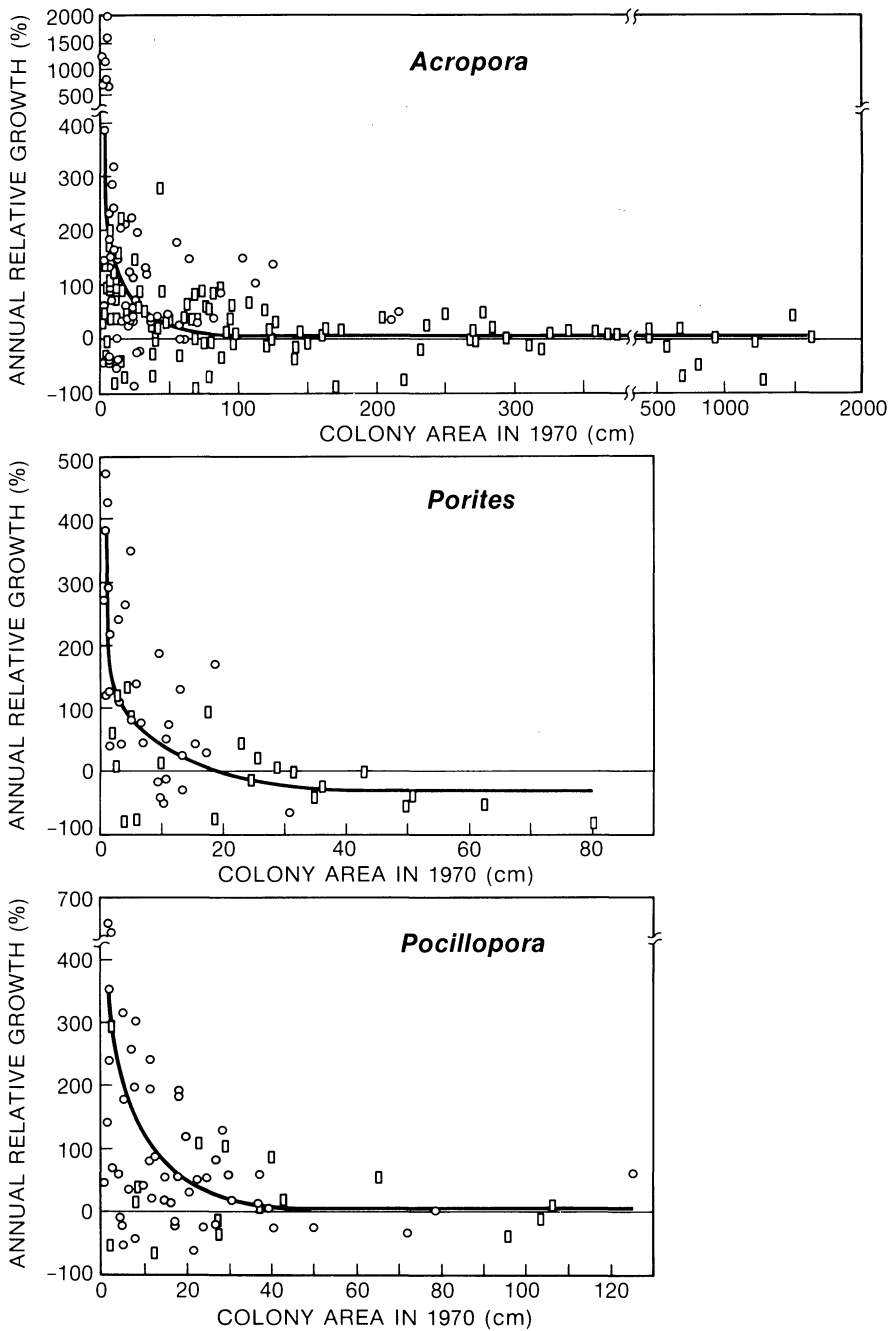


FIG. 1.—Net relative growth of colonies from 1970 to 1971 versus initial colony area. Age of colonies in 1970: *squares*, older than 4.5 yr; *circles*, younger than 4.5 yr. Note that small corals can be young or old. Correlation coefficients (r^2) of nonlinear regression ($y = ax^{-b} + c$) fitted to all points are 0.76 for *Acropora*, 0.89 for *Porites*, and 0.71 for *Pocillopora*. In all cases, $P < 0.001$.

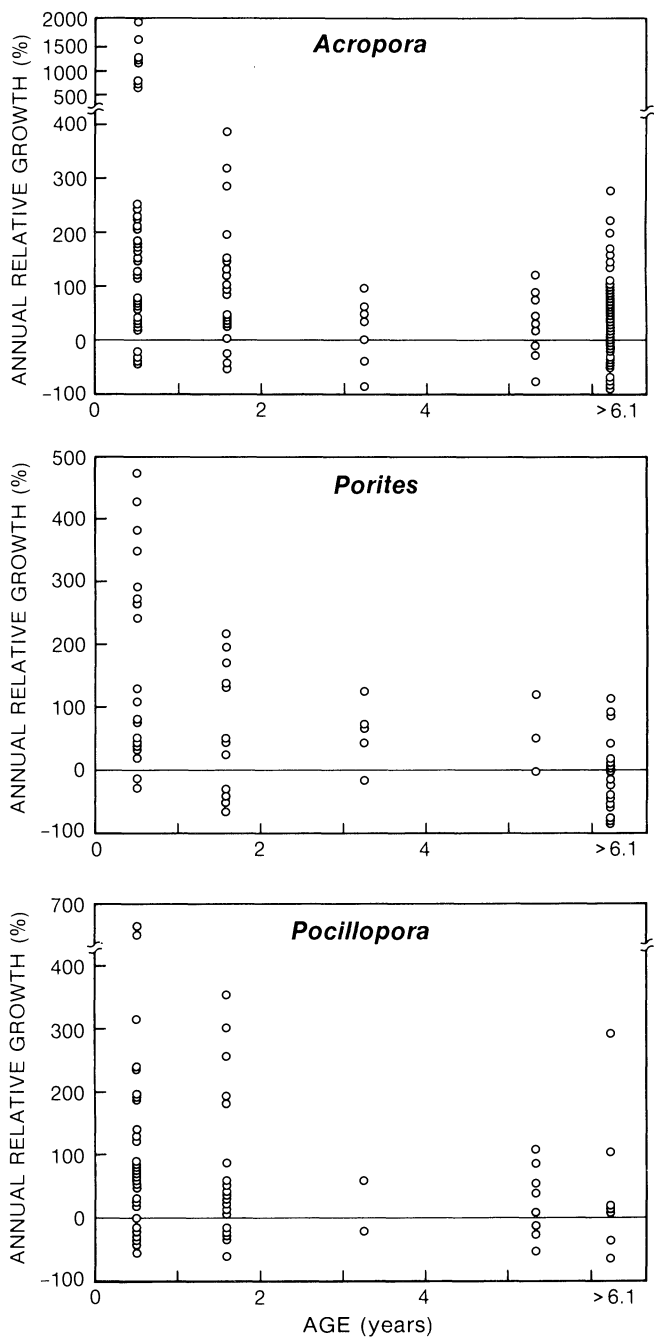


FIG. 2.—Net relative growth of colonies from 1970 to 1971 versus age (length of time since recruitment) in 1970. Points for each of the four younger age classes are plotted at the midpoints of the census intervals. (See the text for analysis.)

TABLE 3
ANALYSIS OF DISTRIBUTION OF GROWTH RATES VERSUS AGE

GROWTH CLASS	<i>Acropora</i> ***		<i>Porites</i> *		<i>Pocillopora</i> †	
	Young (%)	Old (%)	Young (%)	Old (%)	Young (%)	Old (%)
-99%-0%	14	29	24	55	24	38
>0%-100%	38	61	37	36	45	44
>100%	48	10	39	9	31	19
N	63	110	46	22	51	16

NOTE.—Growth class -99%-0% signifies negative or zero percent relative growth.

* $\chi^2 = 8.7$, $P < 0.05$.

*** $\chi^2 = 31.6$, $P < 0.001$.

† $\chi^2 = 1.6$, not significant.

(greater than or less than 4.5 yr, respectively) in two of the three taxa by comparing age distributions among three growth classes (-99%-0%, >0%-100%, and >100%). Overall, young colonies of *Acropora* and *Porites* were four times more likely than older colonies to increase their area by more than 100%, whereas older colonies were twice as likely to shrink (table 3). *Pocillopora* colonies showed the same trends, but the age effect was not significant.

Age Effects Independent of Size

Although these data indicate that initial size and age each influence coral growth, their effects may not have acted independently. The results reflect, in part, the same demographic trends during the ontogeny of a coral because older colonies also tend (albeit with many exceptions) to be larger (fig. 1). In addition, there may also be some interactive effect of size and age. An analysis addressing all age and size classes is not possible, since too few of the younger corals have had sufficient time to grow into larger size classes (fig. 1). This unbalanced situation probably applies to most organisms and may account for the absence of such analyses in the literature.

As an alternative, we can analyze for the effects of age independently of the effects of size by examining corals of mixed ages within a restricted range of sizes. The dynamics of corals older and younger than 4.5 yr were examined as before, but we excluded larger corals that tended to have insufficient numbers of young colonies. Thus, we excluded from the analysis colonies greater than 20 cm² for *Porites*, the slowest grower, and greater than 50 cm² for *Acropora* and *Pocillopora*, and we examined more closely 298 "small" colonies that were both young and old (table 4). To test the hypothesis that similarly sized corals of different ages are equally likely to survive and grow, we looked for statistical interactions between the fate of a small colony and its age, by partitioning variance using a log-linear model (Bishop et al. 1980). The three main effects were taxon (*Acropora*, *Porites*, *Pocillopora*), age (less than or greater than 4.5 yr), and fate between 1970 and 1971 (died, grew 100% or less, grew more than 100%).

The age-by-fate interaction was significant (table 4), indicating that the fate of

TABLE 4

ANALYSIS OF AGE EFFECTS ON THE FATE (SURVIVAL AND GROWTH) OF SMALL COLONIES

A. Percentage of Colonies in Each Fate Category

FATE	<i>Acropora</i>		<i>Porites</i>		<i>Pocillopora</i>		ALL SPECIES	
	Young	Old	Young	Old	Young	Old	Young	Old
Died	30	39	38	55	19	37	29	42
Grew $\leq$ 100%	36	46	29	36	53	47	39	44
Grew $>$ 100%	34	15	33	9	28	16	32	14
Total number	73	74	52	22	58	19	183	115

B. Analysis of Interactions

Interactions	df	χ^2
Age by fate	2	13.5**
Taxon by fate	4	8.6
Age by taxon by fate	4	1.8

NOTE.—Analysis includes only colonies smaller than 50 cm² for *Acropora* and *Pocillopora* and smaller than 20 cm² for *Porites*.

** $P < 0.01$.

small colonies was not independent of their age. The nonsignificant three-way interaction suggests that all taxa behaved in the same way with respect to age. The taxon-by-fate effect was also not significant, implying that small colonies of the three taxa are demographically very similar. (The three main effects and the remaining two-way interaction of species by age were all significant; however, these results reflect only different sample sizes.) Table 4 indicates that the greatest difference between age groups was in the frequency of fast growth. Young small colonies were twice as likely to double their area as old colonies of the same size (fig. 1, table 4). In addition, older small colonies also died at a much higher rate in each of the three taxa.

DISCUSSION

The simultaneous classification of corals into age and size categories (table 1) shows that colonies of the same age often vary greatly in size (and vice versa). Among young colonies, the observed range in sizes was caused mostly by differences in growth rate (fig. 1). Note, however, that there are proportionately fewer young colonies among bigger size classes, since only older colonies had sufficient time to grow very large. The older corals also differed in size, but in their case the mechanism was seldom differential rates of growth. More commonly, they ranged in size because many had shrunk by varying amounts as a result of partial mortality (injury) or fission (fig. 1).

We have demonstrated that size and age each influence the relative growth of corals. Significantly, the substantial difference in growth rates between small young colonies and small old ones (table 4) could not have been predicted from measurements based on size alone (fig. 1) or age alone (fig. 2). The biological

mechanisms causing the differences in net growth and survival rates among old and young small colonies are not known, but previous studies indicate they may be physiological, genetic, or environmental. One possible influence on growth is the redirecting of reserves toward sexual reproduction among older, postreproductive colonies (Loya 1985). Older corals that suffer partial mortality or fission often retain the ability to breed even after they have shrunk below the minimum size required for younger colonies to reproduce (Kojis and Quinn 1985; Szmant-Froelich 1985). Similarly, old shrunken colonies may translocate resources toward regeneration of damaged tissues before resuming new growth (Tardent 1963; Bak 1983; Wahle 1983). The accumulation of commensals, parasites, and skeletal borers (Abele and Patton 1976; Highsmith 1981) or of deleterious somatic mutations (Whitham and Slobodchikoff 1981; Gill and Halverson 1984) in older colonies is also a plausible mechanism for causing a reduction in growth rate or an increase in mortality compared with younger colonies of the same size.

We have found only a handful of other examples (three studies of short-lived herbaceous plants and one each of the moose and a clam) in which the consequences of size and age have been measured together (Werner 1975; Van der Meijden and Van der Waals-Kovi 1979; Gross 1981; Saether and Haagenrud 1983; Peterson 1986). As we have shown here for corals, the size of the plants and the clam were consistently more highly correlated with the likelihood of survival, growth, or reproduction, although in some cases age also had significant effects even within single size classes (see analysis in Caswell 1986). By contrast, fecundity of the moose (the only organism studied so far whose growth is strictly determinate) depends on individual age alone, regardless of size (Saether and Haagenrud 1983; Caswell 1986). Other, indirect evidence for interactions of size and age is provided by studies showing that demographic fates depend on growth rates, since variable growth rates will also result in a mixture of ages within a single size class (Young 1985).

Most predictive population models are based on either age or size, but seldom on both. The effects of size are clearly much greater for clonal organisms, whereas the opposite may be true for mammals, such as the moose. The addition of age to a size-dependent model of coral dynamics (Hughes 1984) would probably not affect its accuracy as much as, say, the incorporation of density dependence or temporal variation, since crowding and episodic storms are significant phenomena on coral reefs. However, if it can be demonstrated that size and age each have important demographic effects, any model incorporating only one may be seriously deficient. Although evidence indicates that both can be significant, they have only very rarely been incorporated together in predictive population models, either by simultaneously analyzing size-dependent and age-dependent vital rates (Sinko and Streifer 1969; Law 1983) or by incorporating variable stage durations and time lags (Hubbell and Werner 1979; Nisbet and Gurney 1983). Although such models are of necessity more complex, empirical evidence indicates that they may provide a better understanding of mechanisms as well as more-accurate predictions of population dynamics, especially for organisms whose mode of development results in a significant range of size within age classes.

SUMMARY

Much of the variability in the dynamics of individuals in a population is due to differences in age or size. We simultaneously classified colonies of three coral taxa into age and size classes and found that virtually every size class contained colonies that were both young and old. Rates of growth and survival of colonies were measured in order to examine, for the first time for an animal population, the influence of age on growth and survival regardless of size, and vice versa. Size effects were stronger, but the age of the colony also had a significant impact on growth rates of the colony in two of the three taxa. Age effects independent of size were also found. Specifically, old small colonies died more often and grew more slowly than younger corals of the same size. Although biologists have traditionally viewed size and age as alternative ways of classifying populations, models based on simultaneous analyses of size and age may provide greater insight into the dynamics of organisms with highly variable rates of development.

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